Assignment 12 – Project - Foreground and Background Processes in RHEL 9

To understand and work with foreground and background processes in a Red Hat Enterprise Linux 9 environment.

- Red Hat Enterprise Linux 9 installed (either on a physical machine or a virtual machine).
- Access to a terminal or command line interface.

Questions for this assignment

1. Define the below task with answer:

Task 1: Running Processes in the Foreground

2. Define the below task with answer:

Task 2: Running Processes in the Background

3. Define the below task with answer:

Task 3: Bringing Background Processes to the Foreground

Solution by the Instructor:

1. Define the below task with answer:

   Task 1: Running Processes in the Foreground

   1.1. Open a terminal and run a simple command, such as ls or pwd, in the foreground.

   ls

   1.2. Observe how the terminal is occupied by the running process. Explain what happens when a process runs in the foreground.

   When a process runs in the foreground, it occupies the terminal, and the command prompt is not available until the process completes or is interrupted (e.g., by pressing Ctrl+C to terminate the process).

2. Define the below task with answer:

   Task 2: Running Processes in the Background

2.1. Open a terminal and run a command in the background using the & symbol.

Answer 2.1:

sleep 10 &

This command runs the sleep 10 command in the background, allowing you to regain control of the terminal.

2.2. Use the jobs command to list background jobs.

Answer 2.2:

jobs

Explain the output of the jobs command. What information does it provide about background jobs?

The jobs command lists the background jobs associated with the current shell session. It displays the job ID, status, and command associated with each background job.

3. Define the below task with answer:

Task 3: Bringing Background Processes to the Foreground

3.1. Use the fg command to bring a background process to the foreground.

Answer 3.1:

fg %1

Replace %1 with the job ID obtained from the jobs command.

3.2. Observe how the background process is now running in the foreground. Explain any differences you observe compared to running the process directly in the foreground.

Bringing a background process to the foreground allows it to occupy the terminal, similar to running a process directly in the foreground. However, the key difference is that the process was initially started in the background, allowing you to continue working in the terminal while the process runs.